

BEGIN

Let vehicle-type, is-member, is-disabled, station-avail: CHAR
Let soc, req-level, duration, time: INT
Let req-charging, rate, base-charging-cost, charging-disc: FLOAT
Let base-parking-cost, parking-disc, final-payable: FLOAT
Let priority, time-status, warning: STRING

PRINT "Enter Vehicle Type (E for Electric, H for Hybrid): "
INPUT vehicle-type
PRINT "Enter Current Battery Level (SOC (%)): "
INPUT soc
PRINT "Enter required charging level (%): "
INPUT req-level
PRINT "Enter Expected parking duration (hours): "
INPUT duration
PRINT "Enter Current Time (24 hour Format): "
INPUT time
PRINT "Is Customer a Member (Y/N): "
INPUT is-member
PRINT "Is Disabled-Person Priority? (Y/N): "
INPUT is-disabled
PRINT "Is Charging Station Available? (Y/N): "
INPUT station-avail.

SET base-charging-cost = 0.0

SET charging-disc = 0.0

Date: _____

if (station-avail == 'N') then

if (vehicle-type == 'H') then

PRINT "Charging unavailable - Parking only"

else

PRINT "No charging slot available"

endif

else if (vehicle-type == 'H' AND soc >= 40) then

PRINT "Vehicle does not qualify for EV charging."

else

req-charging = req-level - soc

if (req-charging <= 0) then

PRINT "No charging required"

else

if (soc <= 15 AND req-level >= 80) then

priority = "Emergency Charging Priority"

else if (is-disabled == 'Y' OR is-member == 'Y')

AND (soc <= 30) then

priority = "Priority Charging"

end if

if (time >= 17) AND (time <= 22) then

time-status = "Peak"

rate = 50.0

else

time-status = "Off-Peak"

rate = 35.0

endif

$\text{base_charging_cost} = \text{req_charging} * \text{rate}$

if (priority == "Emergency Charging Priority") then
 charging-disc = 0.0

else if (is-member == 'Y') then

 if (time-status == "Off-Peak") then

 charging-disc = base-charging-cost * 0.20

 else

 charging-disc = base-charging-cost * 0.10

 endif

endif

endif

endif

if (is-disabled == 'Y') then

 parking-disc = base-parking-cost

else if (is-member == 'Y') then

 parking-disc = base-parking-cost * 0.20

else

 parking-disc = 0.0

endif

$\text{final_payable} = (\text{base_charging_cost} - \text{charging_disc}) +$
 $(\text{base_parking_cost} - \text{parking_disc})$

if (duration > 8) then

 warning = "Long stay warning: Please relocate your vehicle"

else

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warning = "Standard parking duration"  
endif
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PRINT "-- SMART EV SYSTEM BILL SUMMARY --"  
PRINT "Vehicle Type: ", vehicle-type  
PRINT "Current SOC: ", soc, "% | Required: ", req-level, "%"  
PRINT "Priority: ", priority  
PRINT "Time Status: ", time-status  
PRINT "Charging Cost: Rs. ", (base-charging-cost - charging-disc)  
PRINT "Parking Cost: Rs. ", (base-parking-cost - parking-disc)  
PRINT "Total Discount: Rs. ", (charging-disc + parking-disc)  
PRINT "Final Payable Amount: Rs. ", final-payable  
PRINT "Message: ", warning  
END.
```